Eleven Knobs, One Mechanism, Zero Adoptions: A Measurement Improvement Worth More Than the Levers It Killed

Claude Opus 5 (deep-analysis author), OPHIS [1]

Abstract

We report six blocks that adopted nothing, and argue that the session's measurement work is worth more than any lever it tested. Twelve interventions were run against the walltime_5min_h200 frame: eleven were knobs (existing environment flags) and one was a mechanism (new code). All twelve failed. The session's own measurements explain why the eleven could not have succeeded: the intrinsic quality effect of a lever, isolated as the residual against a fitted token-throughput law, is ≈ 0.0007, while throughput effects are $\approx$ 0.01 and the adoption gate is 0.004072. No knob in this architecture can clear that gate, and we launched six more knobs after establishing this. The positive contribution is a variance reduction. Contention on the shared host perturbs step count, step count determines tokens, and tokens determine val_bpb through a law we measured; that path is therefore predictable and subtractable rather than something to average away. Used as a control variate on an intervention that provably cannot affect throughput, it cut the paired-delta standard deviation 6.5×, from 0.001469 to 0.000226, moving 5/6 same-sign to 6/6 (t from -2.53 to -6.51). It then overturned three prior readings, including two of our own apparent wins: a lever that looked like -0.0016 became a null of -0.000125, and an effect attributed to Muon momentum-continuity was reassigned to the warmdown ratio. Since the recorded single-seed noise floor is $\sigma = 0.00061$, i.e. 72% of the gate, this changes what the frame can resolve at all: at sd = 0.000226 the standard error at n=3 is 0.00013. Finally, the one mechanism attempted — a static-shape cu_seqlens to delete a per-step host synchronization from the adopted attention path — was verified byte-identical over 200 randomized trials and then rejected by the kernel with an illegal memory access, yielding a registered capability gap and a sharper lesson: content equivalence and kernel-contract compatibility are separate properties, and only the first is testable off-GPU.

1. Next Experiments

This section leads by protocol. Every proposal below is a MECHANISM; the plateau rule now in docs/EXPERIMENT_WORKFLOW.md forbids knob experiments until a mechanism clears the noise band, and §2 is the worked justification.

1.1. M1 — FlashAttention-4 varlen with explicit num_seqs

Mechanism. Block 23 established that fa3 varlen infers its sequence count from cu_seqlens.numel()-1, so a fixed-capacity padded buffer schedules work for sequences whose rows do not exist. A kernel accepting num_seqs separately from the buffer length admits the static-shape construction unchanged, deleting a per-step cudaStreamSynchronize and removing the unbacked symbolic shape that hard-disables CUDA-graph capture. FA4 is shipped (flash-attn-4 v4.0.0b23) and Hopper-supported; PyTorch's published H200 FlexAttention+FA4 figures give 1.41–1.89× forward and 1.48–2.01× backward on document masking specifically, our exact adopted lever. Predicted: gate is reachable at +7.0% tokens ($0.06 \ln r = 0.004072$). Tempered strongly by our own profile: attention is not the measured bottleneck, so the realized share is likely well below the kernel-level speedup. Falsifier: if end-to-end ms/step improves by <2%, attention is off the critical path at this span and the direction is exhausted — stop, do not spend seeds.

[1] Autoresearch reimplementation campaign. Correspondence to: Claude Opus 5 (OPHIS) <baiyuzhu@mit.edu>.

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Risk: beta package, requires CuTeDSL and a recent toolchain; the install itself may destabilize the frozen harness. Install into a separate environment and reconcile before any scored run.

1.2. M2 — Narrow the n-gram value embedding ($384 \to 192$)

Mechanism. The refuted lever was table rows; the untested dimension is width. The dominant batch-independent cost is plausibly embedding_dense_backward zero-filling a $3.22\text{M} \times 384$ gradient per table per step ($\approx$2.5 GB bf16), corroborated by the recorded NGRAM_TABLE_MULT= $512 \to$ OOM. Halving width halves that memset, the scatter, and the optimizer traffic. Why it is the critical path: it is the shared unblocker for two levers that both died on VRAM this session — CUDA-graph capture ($\approx$25 GB of graph pools) and $B \geq 144$ (CUDA OOM). Neither can be retried without it. Risk: HIGH on quality — this cuts capacity of the one module carrying the data-limitation signal. Treat as a capacity experiment, not free throughput.

1.3. M3 — Re-analyze the session's nulls under the control variate

Mechanism. If the correction of §3 reduces resolvable effect size by 6.5×, then levers previously recorded as null may carry real intrinsic effects that the raw frame could not see. This is a zero-GPU re-analysis of data already collected. Falsifier: if no recorded null moves outside ± 0.0007 intrinsic, the nulls are genuine and the control variate buys sensitivity without changing any verdict. Caveat: valid only for interventions that cannot themselves change throughput; otherwise the correction removes signal along with noise and the decomposition is not identified.

2. Why the Eleven Knobs Could Not Have Worked

Break-even for a mechanism costing s step time is $0.06 \cdot (-\ln(1-s))$: $1\% \to 0.00060$, $5\% \to 0.00308$. Against a measured intrinsic ceiling of ≈ 0.0007 and a gate of 0.004072, the admissible set is {zero-cost, negative-cost} and the achievable intrinsic gain is an order of magnitude short. We derived this ratio mid-session and then launched six further knobs, which is the failure the plateau rule now blocks. The classification — eleven knobs, one mechanism, zero adoptions — is recorded because the drift was invisible from inside: each null felt like ruling something out.

3. The Control Variate

Applied to MUON_MOMENTUM_CONTINUOUS, which is host-side scalar arithmetic and provably cannot change throughput, every step-count difference between arms is contention. Subtracting the predicted component gives:

	mean	sd	sign	t
raw	-0.001518	0.001469	5/6	-2.53
intrinsic	-0.000600	**0.000226**	**6/6**	-6.51

Three prior readings were then overturned: MUON_NS_STEPS= 4 from ambiguous to intrinsically worse (+0.0022, matching the theoretical cost of a dropped Newton–Schulz iteration), NGRAM_VE_LR_SCALE= 0.5 from an apparent -0.0016 win to a null (-0.000125), and the momentum effect reassigned to the warmdown ratio by a 2×2 factorial. Two of the three were our own claims.

4. A Mechanism Rejected by Its Kernel

The static-shape cu_seqlens construction was verified byte-identical to the torch.nonzero version over 200 randomized trials before any GPU time, then died on all three treatment arms with an illegal memory access while all three controls completed normally — confirming both the diagnosis and that flag-gating protected the adopted path. The lesson generalizes: a local equivalence test establishes content correctness and is silent on contract compatibility. For kernel-facing changes the second property is the binding one and cannot be tested off-device.

5. Conclusion

Six blocks, twelve interventions, zero adoptions, and a best configuration still -0.0033 from the pre-session baseline — inside the gate, i.e. not an improvement. The durable output is a scoped law validated out of sample, a 6.5× measurement improvement that has already corrected three conclusions, a registered capability gap that converts a vague kernel aspiration into a specific requirement, and two protocol rules that make the session's own failure modes structurally harder to repeat. We record this because a campaign that reports only its wins learns slowly, and because the most valuable artifact this session produced was a better instrument rather than a better model.